

Partial Differential Equations (PDE)

An equation involving partial differential coefficients of two or more independent variables is called a partial differential equation.

Ex ; (i) $\frac{\partial u}{\partial x} = k \frac{\partial u}{\partial y}$ ——— (1)

(ii) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$ ——— (2)

are partial differential equations in eq (1) u is function of two independent variables x and y whereas in equation (2) it is a function of 3 independent variables x, y and z .

⇒ Order of Partial Differential Equation

The order of a P.D.E is the order of the highest partial differential coefficient which occurs in a term.

Example ;

1. $\frac{\partial u}{\partial x} = k \frac{\partial u}{\partial y}$

Order = one.

2. $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$

Order = two.

⇒ Degree of Partial Differential Equation

The degree of a PDE is the power of the highest partial differential coefficient which occurs in, after the differential equation has been made free from radicals and fractions for as the differential coefficient are concerned.

Ex ; $\left(\frac{\partial^2 u}{\partial x^2}\right)^2 + 2x^{3/2} \frac{\partial u}{\partial y} + 5u = 0$

Order = 2 , Degree = 2.

$\Rightarrow$ Linear and Non-Linear P.D.E

A PDE is said to be linear if the dependent variable u and its partial differential occurs only in the first degree, and no product of u and/or its partial differential coefficient occurs in the equation.

A PDE which is not linear is called non-linear.

$\Rightarrow$ Formation of P.D.E

If z is a function of independent variables x and y .

$$\frac{\partial z}{\partial x} = p,$$

$$\frac{\partial^2 z}{\partial x^2} = r$$

$$\frac{\partial z}{\partial y} = q,$$

$$\frac{\partial^2 z}{\partial y^2} = t$$

$$\frac{\partial^2 z}{\partial x \partial y} = s$$

$\Rightarrow$ Formation of PDE by elimination of arbitrary constant.

Let a function

$$f(x, y, z, a, b) = 0 \quad \text{--- (1)}$$

where z is a function of two independent variables x and y and a, b are arbitrary constant.

Partially differentiating eq (1) w.r.t x and y respectively

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial z} \cdot \frac{\partial z}{\partial x} = 0$$

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial z} \cdot p = 0 \quad \text{--- (2)}$$

$$\frac{\partial f}{\partial y} + \frac{\partial f}{\partial z} \cdot \frac{\partial z}{\partial y} = 0$$

$$\frac{\partial f}{\partial y} + \frac{\partial f}{\partial z} \cdot q = 0 \quad \text{--- (3)}$$

Eliminating 'a' and 'b' from equation of the form.

$F(x, y, z, p, q) = 0$ which is a partial differential equation of order 1.

Q. Find the PDE by eliminating of a and b from

$$z = ax + by + ab \quad \text{--- (1)}$$

Partially differentiating (1) w.r.t x

$$\therefore \frac{\partial z}{\partial x} = a = p \quad \text{--- (2)}$$

Now, partially differentiating (1) w.r.t y

$$\frac{\partial z}{\partial y} = b = q \quad \text{--- (3)}$$

$\therefore$ From eq (1), (2) and (3). We get:

$$z = px + qy + pq \quad \text{Ans.}$$

Q. Find PDE by the elimination of a and b.

$$z = (x+a)(y+b) \quad \text{--- (1)}$$

Partially differentiating (1) w.r.t x

$$\therefore \frac{\partial z}{\partial x} = y+b = p \quad \text{--- (2)}$$

Now, partial differentiating (1) w.r.t y

$$\frac{\partial z}{\partial y} = x+a = q \quad \text{--- (3)}$$

From (1), (2) and (3) we get

$$z = pq \quad \text{Ans.}$$

⇒ Formation of PDE by elimination of arbitrary function.
 Let u and v be two functions of x, y, z which are connected by a relation of the form

$$f(u, v) = 0 \text{ ————— (1)}$$

We treat z as the dependent variable and x and y as the independent variables.

$$\frac{\partial y}{\partial x} = 0 \text{ and } \frac{\partial x}{\partial y} = 0 \text{ ————}$$

Differentiating eq (1) w.r.t x and y

We get

$$\frac{\partial f}{\partial u} \left(\frac{\partial u}{\partial x} \times \frac{\partial x}{\partial x} + \frac{\partial u}{\partial y} \times \frac{\partial y}{\partial x} + \frac{\partial u}{\partial z} \cdot \frac{\partial z}{\partial x} \right) +$$

$$\frac{\partial f}{\partial v} \left(\frac{\partial v}{\partial x} \cdot \frac{\partial x}{\partial x} + \frac{\partial v}{\partial y} \cdot \frac{\partial y}{\partial x} + \frac{\partial v}{\partial z} \cdot \frac{\partial z}{\partial x} \right) = 0$$

$$\text{i.e. } \frac{\partial f}{\partial u} \left(\frac{\partial u}{\partial x} + 0 + p \frac{\partial u}{\partial z} \right) + \frac{\partial f}{\partial v} \left(\frac{\partial v}{\partial x} + p \frac{\partial v}{\partial z} \right) = 0 \text{ — (2)}$$

Similarly differentiating eq (1) w.r.t y we get

$$\frac{\partial f}{\partial u} \left(\frac{\partial u}{\partial y} + q \frac{\partial u}{\partial z} \right) + \frac{\partial f}{\partial v} \left(\frac{\partial v}{\partial y} + q \frac{\partial v}{\partial z} \right) = 0 \text{ — (3)}$$

Eliminating $\frac{\partial f}{\partial u}$ and $\frac{\partial f}{\partial v}$ in eq (2) and (3)

$$\begin{vmatrix} \frac{\partial u}{\partial x} + p \frac{\partial u}{\partial z} & \frac{\partial v}{\partial x} + p \frac{\partial v}{\partial z} \\ \frac{\partial u}{\partial y} + q \frac{\partial u}{\partial z} & \frac{\partial v}{\partial y} + q \frac{\partial v}{\partial z} \end{vmatrix} = 0$$

$$\left(\frac{\partial u}{\partial x} + p \frac{\partial u}{\partial z}\right) \left(\frac{\partial v}{\partial y} + q \frac{\partial v}{\partial z}\right) - \left(\frac{\partial v}{\partial x} + p \frac{\partial v}{\partial z}\right) \left(\frac{\partial u}{\partial y} + q \frac{\partial u}{\partial z}\right) = 0$$

i.e

$$\left(\frac{\partial u}{\partial y} \cdot \frac{\partial v}{\partial z} - \frac{\partial u}{\partial z} \frac{\partial v}{\partial y}\right) p + \left(\frac{\partial u}{\partial x} \frac{\partial v}{\partial z} - \frac{\partial u}{\partial z} \frac{\partial v}{\partial x}\right) q = \left(\frac{\partial u}{\partial x} \cdot \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \cdot \frac{\partial v}{\partial x}\right)$$

i.e

$$\frac{\partial(u,v)}{\partial(y,z)} p + \frac{\partial(u,v)}{\partial(z,x)} q = \frac{\partial(u,v)}{\partial(x,y)}$$

i.e

$$Pp + Qq = R.$$

Q. Form a partial differential equation eliminating arbitrary functions.

$$z = f(x+2y) \quad \text{--- (1)}$$

Partially differentiating (1) w.r.t x

$$p = \frac{\partial z}{\partial x} = f'(x+2y) \quad \text{--- (2)}$$

Similarly differentiating (1) w.r.t y

$$q = \frac{\partial z}{\partial y} = f'(x+2y) \times 2 \quad \text{--- (3)}$$

From (2) and (3) we get

$$\frac{p}{q} = \frac{f'(x+2y)}{2x f'(x+2y)} = \frac{1}{2}.$$

$$q = 2p \quad \text{Ans.}$$

Q. $z = x + y + f(xy)$

$$\therefore p = \frac{\partial z}{\partial x} = 1 + f'(xy) \cdot y \quad \text{--- (1)}$$

$$q = \frac{\partial z}{\partial y} = 1 + f'(xy) \cdot x \quad \text{--- (2)}$$

From (1) and (2)

$$\therefore \begin{array}{l} xp = x + f'(xy) \cdot xy \\ yq = y + f'(xy) \cdot xy \end{array}$$

$$xp - yq = x - y \quad \text{Ans.}$$

$\Rightarrow$ Lagrange's Linear P.D.E

The PDE of the form $Pp + Qq = R$ where P , Q and R are functions of x, y, z is the standard form of the linear P.D.E of the first order.

It is called Lagrange's Linear equation.

$\Rightarrow$ Solution of Lagrange's Linear Equation.

Q. $p + q + 1 = 0$

Here Lagrange's Auxiliary equation are

$$\frac{dx}{1} = \frac{dy}{1} = \frac{dz}{1}$$

Taking (1) and (2) fraction

$$\begin{array}{l} dx = dy \\ x = y + C_1 \\ x - y = C_1 \end{array}$$

Again taking (2) and (3) fractions

$$dy = dz$$

$$y = z + C_2$$

$$y - z = C_2$$

∴ The required solution

$$\phi(C_1, C_2) = 0$$

$$\Rightarrow \phi(x-y, y-z) = 0 \quad \text{Ans.}$$

a $2p + 3q = 1$

∴ The Lagrange's auxiliary equation are ;

$$\frac{\partial x}{2} = \frac{\partial y}{3} = \frac{\partial z}{1}$$

Taking (1) and (2) fractions

$$3 \partial x = 2 \partial y$$

$$3x = 2y + C_1$$

$$3x - 2y = C_1$$

Again, taking (2) and (3) fractions

$$\frac{\partial y}{3} = \frac{\partial z}{1}$$

$$\partial y = 3 \partial z$$

$$y = 3z + C_2$$

$$y - 3z = C_2$$

∴ The required solution is ;

$$\phi(C_1, C_2) = 0$$

$$\Rightarrow \phi(3x - 2y, y - 3z) = 0 \quad \text{Ans.}$$

Q. Solve: $xz p + yz q = xy$

Lagrange's Auxiliary equation:

$$\frac{dx}{p} = \frac{dy}{q} = \frac{dz}{R}$$

$$\therefore \frac{dx}{xz} = \frac{dy}{yz} = \frac{dz}{xy}$$

$\therefore$ Taking fraction (1) and (2)

$$\frac{dx}{xz} = \frac{dy}{yz}$$

$\therefore$ Integrating both sides

$$\int \frac{dx}{x} = \int \frac{dy}{y}$$

$$\log x = \log y + \log c_1$$

$$\log x = \log c_1 y$$

$$x = c_1 y$$

$$c_1 = \frac{x}{y}$$

Again taking fraction (2) and (3)

$$\frac{dy}{yz} = \frac{dz}{xy}$$

Putting $x = y c_1$

$$\therefore \frac{dy}{z} = \frac{dz}{c_1 y}$$

$$\Rightarrow c_1 y dy = z dz$$

Integrating both sides

$$c_1 \times \frac{y^2}{2} = \frac{z^2}{2} + c_2$$

$$\Rightarrow \frac{x}{y} \times \frac{y^2}{2} = \frac{z^2}{2} + c_2 \Rightarrow xy - z^2 = c_2 \text{ Ans.}$$

$$\phi(c_1, c_2) = 0$$

$$\therefore \phi\left(\frac{x}{y}, xy - z^2\right) = 0 \quad \text{Ans.}$$

Q. $p - q = \frac{z}{x+y}$

Lagrange's Auxiliary equation;

$$\frac{dx}{1} = \frac{dy}{-1} = \frac{dz}{z/(x+y)}$$

Taking fractions (1) and (2),

$$\frac{dx}{1} = -\frac{dy}{1}$$

Integrating both sides

$$\int dx = \int -dy$$

$$x = -y + c_1$$

$$x + y = c_1$$

Now, again taking fractions (2) and (3)

$$\frac{dy}{-1} = c_1 \frac{dz}{z}$$

Integrating both sides

$$\int -dy = c_1 \int \frac{dz}{z}$$

$$-y = c_1 \ln z + c_2$$

$$c_2 = -y - \frac{\ln z}{(x+y)} \times (x+y)^2 = -y - \ln z (x+y)$$

Hence, the solution is given as;

$$\phi(c_1, c_2) = 0$$

$$\phi(x+y, -y - \ln z (x+y)) = 0 \quad \text{Ans.}$$

Method 3:

Rule 3 for solving $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

If P_1, Q_1, R_1 are functions of x, y, z or constants then from algebra we know that

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} = \frac{P_1 dx + Q_1 dy + R_1 dz}{P_1 P + Q_1 Q + R_1 R}$$

If the multipliers P_1, Q_1, R_1 are so chosen that $P_1 P + Q_1 Q + R_1 R = 0$, then we have

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} = \frac{P_1 dx + Q_1 dy + R_1 dz}{0}$$

$$P_1 dx + Q_1 dy + R_1 dz = 0$$

Q Solve

$$(y-z)p + (z-x)q = (x-y)$$

$$\therefore \frac{dx}{y-z} = \frac{dy}{z-x} = \frac{dz}{x-y} = \frac{P_1 dx + Q_1 dy + R_1 dz}{P_1(y-z) + Q_1(z-x) + R_1(x-y)}$$

$\Rightarrow \frac{dx}{y-z}$ Choosing multipliers

$$P_1 = 1$$

$$Q_1 = 1$$

$$R_1 = 1$$

Taking fractions ① and ④

$$\Rightarrow 1 dx + 1 dy + 1 dz = 0$$

Integrating both sides

$$x + y + z = c_1$$

Again choosing multipliers

$$P_1 = x$$

$$Q_1 = y$$

$$R_1 = z$$

$\Rightarrow$ Taking fractions ① and ④

$$x dx + y dy + z dz = 0$$

Integrating both sides

$$x^2 + y^2 + z^2 = c_2$$

Hence, the solution is given as;

$$f(c_1, c_2) = 0$$

$$\Rightarrow f(x+y+z, x^2+y^2+z^2) = 0 \quad \text{Ans.}$$

Q4:

Solve

$$p + q = x + y + z$$

Lagrange's Auxiliary equations;

$$\frac{1}{dx} = \frac{1}{dy} x$$

$$\frac{dx}{1} = \frac{dy}{1} = \frac{dz}{(x+y+z)}$$

Taking fractions (1) and (2)

$$\frac{dx}{1} = \frac{dy}{1}$$

Integrating both sides

$$x = y + c_1$$

3rd

$$\frac{dx}{p} = \frac{dy}{q} = \frac{dz}{r} = \frac{p_1 dx + q_1 dy + r_1 dz}{p_1(1) + q_1(1) + r_1(x+y+z)}$$

choosing multipliers

$$p_1 = 1$$

$$q_1 = 1$$

$$r_1 = 1$$

We get

$$0 = \frac{d(z+x+y+z)}{z+x+y+z} = y + c_2$$

$\therefore$ on integrating, we get;

$$\log(z+x+y+z) = y + c_2$$

$\Rightarrow$ Solution is given as;

$$f(x-y, \log(z+x+y+z)-y) = 0. \quad \text{Ans.}$$

Q.

Solve

$$(mz - ny)p + (nx - lz)q = ly - mx$$

Lagrange's Auxiliary Equations are given as;

$$\frac{dx}{(mz - ny)} = \frac{dy}{(nx - lz)} = \frac{dz}{(ly - mx)} = \frac{P_1 dx + Q_1 dy + R_1 dz}{P_1(mz - ny) + Q_1(nx - lz) + R_1(ly - mx)}$$

∴ Choosing multipliers

$$P_1 = x$$

$$Q_1 = y$$

$$R_1 = z$$

Taking fractions (1) and (4), we get

$$x dx + y dy + z dz = 0$$

Integrating both sides

$$x^2 + y^2 + z^2 = C_1$$

Again, on choosing multipliers;

$$P_1 = l$$

$$Q_1 = m$$

$$R_1 = n$$

Taking fractions (1) and (4), we get;

$$l dx + m dy + n dz = 0$$

Integrating both sides

$$lx + my + nz = C_2$$

Hence, the solution is given as;

$$\phi(x^2 + y^2 + z^2, lx + my + nz) = 0 \quad \text{Ans.}$$

Q. Solve ;
 $x(y^2 - z^2)P + y(z^2 - x^2)Q = z(x^2 - y^2)R$

Lagrange's Auxiliary equation ;

$$\frac{dx}{x(y^2 - z^2)} = \frac{dy}{y(z^2 - x^2)} = \frac{dz}{z(x^2 - y^2)} = \frac{P_1 dx + Q_1 dy + R_1 dz}{P P_1 + Q Q_1 + R R_1}$$

Choosing multipliers

$$P_1 = x$$

$$Q_1 = y$$

$$R_1 = z$$

Taking fractions ① and ④

$$\Rightarrow \frac{x dx + y dy + z dz}{x^2 + y^2 + z^2} = 0$$

Integrating both sides ;

$$x^2 + y^2 + z^2 = C_1$$

Again choosing multipliers ;

$$P_1 = 1/x$$

$$Q_1 = 1/y$$

$$R_1 = 1/z$$

Taking fractions ① and ④

$$\frac{1/x dx + 1/y dy + 1/z dz}{\ln x + \ln y + \ln z} = 0$$

Integrating both sides

$$\ln x + \ln y + \ln z = \ln C_2$$

$$\ln xyz = \ln C_2$$

$$\Rightarrow xyz = C_2$$

Hence, the solution is given as ;

$$\phi(C_1, C_2) = 0$$

$$\phi(x^2 + y^2 + z^2, xyz) = 0 \quad \text{Ans.}$$

Non-Linear Differential Equation.

→ Charpit's Method

$$\frac{\frac{dp}{\frac{df}{dx} + p \frac{df}{dz}}}{\frac{dq}{\frac{df}{dy} + q \frac{df}{dz}}} = \frac{\frac{dz}{-p \frac{df}{dp} - q \frac{df}{dq}}}{\frac{dx}{-\frac{df}{dp}}} = \frac{\frac{dy}{-\frac{df}{dq}}}{\frac{dz}{-p \frac{df}{dp} - q \frac{df}{dq}}}$$

i.e

$$\frac{dp}{f_x + p f_z} = \frac{q}{f_y + q f_z} = \frac{dz}{-p f_p - q f_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q}$$

These equations are known as Charpit's Auxiliary equations. These equations we known as Charpit's auxiliary equations.

Q.

$$z = pq$$

Find the complete integral of partial differential equation.

Sol. Here the given equation can be written as;

$$f = z - pq = 0 \quad \text{--- (1)}$$

Charpit's Auxiliary equations are;

$$\frac{dp}{f_x + p f_z} = \frac{dq}{f_y + q f_z} = \frac{dz}{-p f_p - q f_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q}$$

$$\frac{dp}{0+p} = \frac{dq}{0+q} = \frac{dz}{-p-q} = \frac{dx}{-1} = \frac{dy}{-1}$$

Taking fractions I. and II

$$\frac{dp}{p} = \frac{dq}{q}$$

Integrating both sides, we get.

$$\log p = \log q + \log a \Rightarrow \log p = \log a q$$

$$\therefore p = aq$$

$$\text{and } a = p/q$$

Substituting this value in equation (1), we get

$$f = z - aq^2 = 0$$

$$\Rightarrow q = \sqrt{\frac{z}{a}}$$

$$\Rightarrow p = \sqrt{za}$$

Putting these values in ;

$$dz = p dx + q dy$$

$$dz = \sqrt{za} dx + \sqrt{\frac{z}{a}} dy$$

$$\Rightarrow \frac{dz}{\sqrt{z}} = \sqrt{a} dx + \frac{dy}{\sqrt{a}}$$

Integrating both sides, we get $\Rightarrow$

$$2z^{1/2} = a^{1/2}x + a^{-1/2}y + b \quad \text{Ans.}$$

Q. Solve

$$zpq = p + q$$

$\therefore$ Here the given equation can be written as ;

$$f = zpq - p - q = 0 \quad \text{--- (1)}$$

Charpit's Auxiliary equations are ;

$$\frac{dp}{f_x + p f_z} = \frac{dq}{f_y + q f_z} = \frac{dz}{-p f_p - q f_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q}$$

$$\therefore \frac{dp}{0 + p(pq)} = \frac{dq}{0 + q(pq)} = \frac{dz}{-p(zq-1) - q(zp-1)} = \frac{dx}{-(zq-1)} = \frac{dy}{-(zp-1)}$$

Taking fractions I and II

$$\frac{dp}{p(pq)} = \frac{dq}{q(pq)}$$

$$\Rightarrow \frac{dp}{p} = \frac{dq}{q}$$

Integrating both sides, we get

$$\log p = \log q + \log a$$

$$\Rightarrow \log p = \log aq$$

$$\Rightarrow p = aq$$

Substituting this value in eq (1), we get

$$\Rightarrow f = zaq^2 - aq - q = 0$$

$$f = zaq^2 - (a+1)q = 0$$

$$\therefore q = \frac{(a+1) \pm \sqrt{(a+1)^2 - 4za(0)}}{2(za)}$$

$$q = \frac{(a+1) \pm (a+1)}{2za}$$

$$\therefore q = \frac{(a+1)}{za} \quad \text{or} \quad q = 0$$

$$\text{If } q = \frac{(a+1)}{za} \Rightarrow p = \frac{(a+1)}{z}$$

Putting values of p and q in

$$dz = p dx + q dy$$

$$dz = \frac{(a+1)}{z} dx + \frac{(a+1)}{az} dy$$

$$z dz = (a+1) dx + \frac{(a+1)}{a} dy$$

Integrating both sides

$$\frac{z^2}{2} = (a+1)x + \frac{(a+1)}{a}y + b$$

$$z^2 = 2\left\{(x+y) + ax + \frac{y}{a} + b\right\} \quad \text{Ans.}$$

Standard Form: First

$$f(p, q) = 0 \quad \text{--- (1)}$$

If the given P.D.E contains only p and q it can be written in the form

$$f(p, q) = 0$$

i.e.

$$f_x = f_y = f_z = 0$$

$\therefore$ The Charpit's equations is given by:

$$\frac{dp}{0} = \frac{dq}{0} = \frac{dz}{-pf_p - qf_q}$$

Taking the first two fractions, we get

$$dp = 0 \text{ and } dq = 0$$

$$p = a \text{ and } q = b \quad \text{--- (2)}$$

Putting these value of p and q in equation (1) obtains

$$f(a, b) = 0$$

Using eq (2), we now have

$$dz = p dx + q dy$$

$$dz = a dx + b dy$$

$$z = ax + by + c$$

This is the required complete integral.

Q. Solve

$$p + q = pq$$

$$f = p + q - pq = 0$$

$$\Rightarrow f(p, q) = 0$$

$$\therefore f_x = f_y = f_z = 0$$

Using First's Standard form, we get

$$\Rightarrow p = a \text{ and } q = b$$

$$\Rightarrow f = a + b - ab = 0$$

$$\Rightarrow a + b = ab$$

$$\Rightarrow a + b(1-a) = 0$$

$$a \neq \frac{a}{b} \neq 1 \quad b = \frac{a}{a-1}$$

The complete integral is given by

$$\Rightarrow z = ax + by + c \Rightarrow z = ax + \frac{a}{a-1}y + c$$

Ans.



Standard Form: Second (Clairaut's Form)

A.P.D.E of the first order is said to be of Clairaut's form if it can be expressed in the form

$$z = px + qy + f(p, q)$$

Q.

$$z = px + qy + pq$$

The given P.D.E is of Clairaut's form

$$\therefore f = z - px - qy - pq = 0$$

$$\Rightarrow f = z - px - qy - f(p, q) = 0$$

It's complete integral is given by;

$$z = ax + by + ab. \quad \underline{\underline{\text{Ans.}}}$$

Only p, q and z present
standard Form (III) :

Consider a PDE of the form $f(p, q, z) = 0$ — (1)
Let

$$u = x + ay \text{ — (2)}$$

Step 1: Replace p and q by $\frac{dz}{du}$ and $a\frac{dz}{du}$ in equation number (1), and solve the resulting O.D.E by suitable method.

Step 2: Put $u = x + ay$ in the solution of O.D.E.

Q. Find the complete integral of
 $q(p^2z + q^2) = 4$

$$\therefore f(p, q, z) = q(p^2z + q^2) - 4 = 0 \text{ — (1)}$$

Let $u = x + ay$
Replacing p and q with $\frac{dz}{du}$ and $a\frac{dz}{du}$ respectively
in (1)

$$\therefore q\left(\left(\frac{dz}{du}\right)^2z + \left(a\frac{dz}{du}\right)^2\right) - 4 = 0$$

$$\left(\frac{dz}{du}\right)^2 = \frac{4}{q(z + a^2)}$$

$$\frac{dz}{du} = \frac{\pm 2}{\sqrt{q(z + a^2)}}$$

$$\therefore \sqrt{z + a^2} dz = \frac{\pm 2}{\sqrt{q}} du$$

Integrating both sides

$$\frac{2}{\sqrt{q}}(z + a^2)^{3/2} = \pm \frac{2}{\sqrt{q}}u + b$$

$$\therefore (z + a^2)^{3/2} = \pm (x + ay) + b$$

Ans.



Q. Find the complete integral of

$$z = pq$$

$$\therefore f(p, q, z) = z - pq = 0 \quad \text{--- (1)}$$

$$\text{Let } u = x + ay$$

Replacing p and q with $\frac{du}{dz}$ and $a \frac{du}{dz}$ respectively in (1)

$$\therefore z - \left(\frac{du}{dz}\right) \left(a \frac{du}{dz}\right) = 0$$

$$\therefore z = a \left(\frac{du}{dz}\right)^2$$

$$\therefore \sqrt{z} dz = \sqrt{a} du$$

Integrating both sides, we get

$$\Rightarrow \frac{2}{3} z^{3/2} = \sqrt{a} u + b$$

$$\therefore \frac{2}{3} z^{3/2} = \sqrt{a} (x + ay) + b$$

$$z - a \left(\frac{dz}{du}\right)^2 = 0$$

$$\Rightarrow \frac{dz}{du} = \sqrt{\frac{z}{a}}$$

$$\therefore \frac{dz}{\sqrt{z}} = \frac{du}{\sqrt{a}}$$

Integrating both sides, we get

$$-2z^{1/2} = \frac{u}{\sqrt{a}} + b$$

$$+2z^{1/2} = \frac{(x + ay)}{\sqrt{a}} + b \quad \text{Ans.}$$

$\Rightarrow$ Standard Form (IV):
Equation of the form:

$$f_1(x, p) = f_2(y, q)$$

Let

$$f(p, q, x, y, z) = f_1(x, p) - f_2(y, q) = 0$$

$$\therefore f_1(x, p) = a \quad \text{and} \quad f_2(y, q) = b$$

and

$$dz = p dx + Q dy$$

$$\Rightarrow p = f_1(x, a) \quad \text{and} \quad Q = f_2(y, b)$$

$$dz = f_1(x, a) dx + f_2(y, b) dy$$

$$z = \int f_1(x, a) dx + \int f_2(y, b) dy$$

Q.

$$x(1+y)p = y(1+x)q$$

$$\therefore \left(\frac{x}{1+x}\right)p = \left(\frac{y}{1+y}\right)q$$

$$\therefore f_1(x, p) = a \quad \text{and} \quad f_2(y, q) = b$$

$$\Rightarrow \left(\frac{x}{1+x}\right)p = a \quad \Rightarrow \left(\frac{y}{1+y}\right)q = b$$

$$\Rightarrow p = \frac{(1+x)a}{x} \quad \text{and} \quad q = \left(\frac{1+y}{y}\right)b$$

$$\therefore dz = f_1(x, a) dx + f_2(y, b) dy$$

$$\int dz = \int \left(\frac{1}{x} + 1\right)a dx + \int \left(\frac{1}{y} + 1\right)b dy$$

$$z = a(\ln x + x) + b(\ln y + y) \quad \text{Ans.}$$